Skills Summary

Interpreting the Solution of a Linear System on a Graph

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading the Solution Point Off a Graph

The model: a line is the set of *all* points whose coordinates make its equation true. Two lines drawn on one grid therefore show two such sets, and the point where they cross belongs to both — that point, written (x, y), is the solution of the system.

Reading it: find the crossing, count across for x, count up for y. Order matters: (across, up).

Always check: substitute the pair into *both* original equations. A point read by eye can sit a square off, and a point that satisfies only one equation is on only one line.

Example: A graph shows $y = x + 3$ and $y = -2x + 9$ crossing two right and five up.
Step 1. A crossing point is the one place both equations can be true at once, so the reading gives a candidate and substitution is what settles it.
Step 2. The candidate is $(2, 5)$. Check: $5 = 2 + 3$ and $5 = -2(2) + 9 = 5$. Both hold, so the solution is **(2, 5)**.

Try it: A graph shows $y = x - 1$ and $y = -x + 7$ crossing four right and three up. Write the solution and check it in both equations.

check: **(4, 3)**

Skill 2 — Solving a System by Graphing

Method: write each equation as $y = mx + b$; plot b on the y-axis; step off the slope ($m =$ rise over run) to a second point; draw the line. Do it for both, then read the crossing.

Standard form is fine too: for $2x + y = 6$, the intercepts are quick — set $x = 0$ to get $(0, 6)$, set $y = 0$ to get $(3, 0)$.

Limits of graphing: it is exact only when the crossing lands on a lattice point. A crossing between grid lines gives an estimate, and you finish with substitution or elimination.

Example: Solve $y = -x + 6$ and $y = 3x - 2$ by graphing.
Step 1. Both are already in slope-intercept form, so each line is fixed by its intercept plus one slope step — no rearranging is needed before plotting.
Step 2. Plot $(0, 6)$ and step down 1, right 1; plot $(0, -2)$ and step up 3, right 1. The lines meet at $(2, 4)$. Check: $-2 + 6 = 4$ and $3(2) - 2 = 4$, so the solution is **(2, 4)**.

Try it: Solve $y = 3x - 4$ and $y = x + 2$ by graphing, then check your point in both equations.

check: **(3, 5)**

Skill 3 — One Solution, None, or Infinitely Many

Compare the slope-intercept forms before you draw:

- different slopes → the lines cross once → **one solution**;
- same slope, different intercept → parallel, never touching → **no solution**;
- same slope *and* same intercept → one line drawn twice → **infinitely many solutions**, every point on the line.

The algebra matches the picture: set the two right sides equal. If the x terms cancel and leave a false sentence such as $-2 = 4$, the lines are parallel. If they cancel and leave a true one such as $6 = 6$, the equations describe the same line.

Example: Classify $y = 3x - 2$ and $y = 3x + 4$.
Step 1. Both slopes are 3 and the intercepts differ, so the lines run parallel and no crossing point can exist — worth knowing before spending time on a graph.
Step 2. Setting the right sides equal gives $3x - 2 = 3x + 4$; the $3x$ cancels and leaves $-2 = 4$, which is false for every x, so no solution. Had the leftover sentence been true, the answer would instead be every point on the line.

Try it: Classify $y = 5x + 1$ and $y = 5x - 6$. How many solutions, and what does the graph look like?

check: none — parallel lines

(!) Watch out: a point that works in one equation is not a solution — it has to work in both. And “no solution” is not the same as “the solution is zero”: $(0, 0)$ is a perfectly good solution when both lines pass through the origin, while parallel lines have no solution point at all.